

Today:

-Start 3D - frames, rigid objects, vectors, angular velocity

Does not deform
Rigid object:

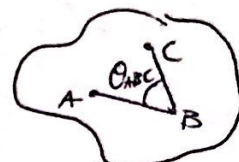
collection of points where $\frac{d}{dt}(d_{ij}) = 0$



↑ distance between i and j

$$\frac{d}{dt}(\theta_{ABC}) = 0$$

↑ any angle between pts
 A, B, C

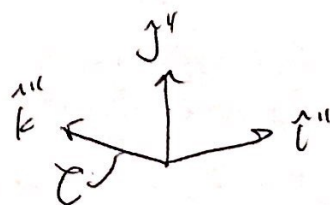
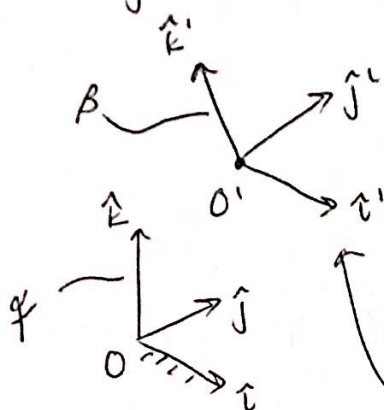


Frame:

• origin + 3 mutually orthogonal unit vectors

• Always a fixed frame

↳ coordinate system, can use laws of mechanics w/ good accuracy



$$\hat{i}' = \hat{e}_i^B = \hat{i}^B$$

different notations ppl use

could be rotating + moving

Vector:

• A geometric thing: an arrow

$$\vec{v} = \vec{v}$$

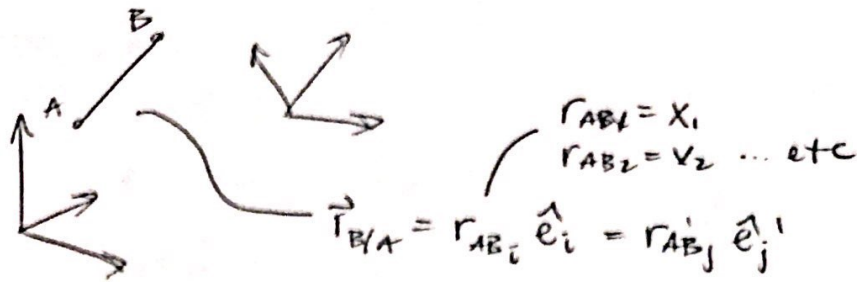
$$v_i \hat{e}_i = v_i' \hat{e}_i'$$

• vector equal to itself, even if components aren't equal

$$[\vec{v}]_\gamma = v_\gamma \neq v_\gamma' = [\vec{v}]_\beta$$

A frame is a rigid object

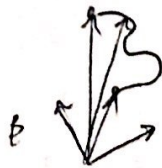
A rigid object is associated with an infinite # of frames] some people call this collection a "frame"



If A is fixed, $\vec{v}_B = \frac{d}{dt}(\vec{r}_{AB}) = \frac{d}{dt}(x_i \hat{e}_i) = \frac{d}{dt}(x_i' \hat{e}_i')$

$$= \dot{x}_i \hat{e}_i = \underbrace{\dot{x}_i' \hat{e}_i' + \dot{x}_i' \hat{e}_i'}_{\frac{d}{dt}(\vec{r})} \quad \underbrace{x_i \hat{e}_i = x_i' \hat{e}_i'}$$

Frame derivative: $\beta \frac{d\vec{r}}{dt} = \dot{r}_i' \hat{e}_i'$



• position vector and coordinate system change in time

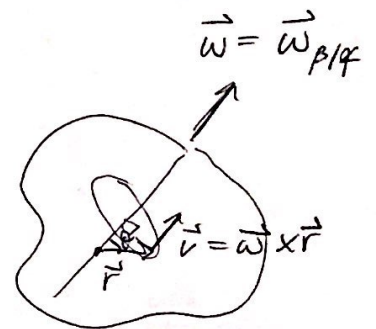
- How does tip of arrow move on graph paper fixed in β ?

$$\beta \frac{d\vec{r}}{dt} = \dot{r}_i' \hat{e}_i'$$

Angular Velocity:

Assume: there exists for any object or frame

$$\vec{\omega} = \vec{\omega}_{\beta/\gamma}$$



$$|\vec{r}| \sin \theta (\vec{\omega}) \hat{\lambda} = \vec{\omega} \times \vec{r}$$

↑
in plane $\perp$ to $\vec{r}$ and $\vec{\omega}$ given by right hand rule

$$\dot{\vec{r}}_{AB} = \vec{\omega}_{\beta/A} \times \vec{r}_{AB}$$

↑
 $\vec{r}_{AB} = \text{Constant in } \beta$

Any vector fixed in β ; $\vec{v}$

$$\dot{\vec{v}} = \vec{\omega}_{\beta/A} \times \vec{v}$$

ex) $\hat{e}_i$ is fixed in β (fixed in moving reference frames)

$$\Rightarrow \dot{\hat{e}}_i = \vec{\omega}_{\beta/A} \times \hat{e}_i$$

↳ " $\dot{\hat{e}}_i$ "

TRANSPORT THEOREM / Q Formula

$$\dot{\vec{r}} = \dot{\vec{r}} + \vec{\omega} \times \vec{r}$$

$$\vec{r}_i \hat{e}_i = r_i \hat{e}_i$$

Back to $\vec{r}$

$$\dot{\vec{r}} = \dot{\vec{r}} = \dot{\vec{r}} = \frac{d\vec{r}}{dt}$$

$$= \frac{d}{dt}(r_i \hat{e}_i) = \underbrace{\dot{r}_i \hat{e}_i}_{\dot{\vec{r}}} + r_i \dot{\hat{e}}_i$$

↳ the frame derivative

For any vector $\vec{Q}$, $\dot{\vec{Q}} = \dot{\vec{Q}} + \vec{\omega} \times \vec{Q}$

Doesn't matter what coordinate system you write $\vec{Q}$ in